Benjamin Diedrichsen

Polyglot Software Engineer

github: https://github.com/bennidi
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The design and construction of software systems - be it small libraries, frameworks or midsize software landscapes - is my job and one of my passions.

This includes many aspects of the software lifecycle:
planning and design, data and domain modeling, prototyping, deployment and production. DevOps is an ongoing learning field...

Polyglot

I write code in various languages using combinations of object oriented and functional paradigm. @Home on the JVM [**Java, Scala, Groovy**], I recently became an enthusiastic user of the JavaScript ecosystem - **node.js, coffeescript, livescript, react.js**

Everything is a model

I enjoy the creative task of deconstructing reality into appropriate models of objects, functions, data structures and information flows. I love writing beautiful and efficient code - git clone my open source projects and have a look.

I also believe in the values of agile methodologies: open and honest communication, collective learning and team collaboration are as important as choosing the right technology stack.

You can hire me to

- design and prototype core modules or support design of software architecture
- fix performance issues in application code or data models
- plan + implement refactorings
- help with code reviews, technology decisions, development cycle and team process

I grant discounts if you

- have a an interesting technology stack
- solve social problems instead of consumer problems

Software projects

12/2016 – 06/2017	**Microservices in polyglot microservice landscape**
Affinitas GmbH *eDarling, ElitePartner*	Java + Spring based microservice for sending push notifications to mobile devices Node.js based microservice family for in-app micropayment.
Technologies	Node.js, CoffeeScript, AWS, koa, Kinesis, Java 8, Spring, RAML, wiremock, JWT, Eureka
Tasks	➢ Migration of legacy Push-Notification feature to standalone microservice based on existing Java + Spring template ➢ Prototyping of Node.js microservice template with CoffeeScript. Integration with Eureka and Spring Config ➢ Design of RESTful API for Micropayment feature. Integration with service landscape (billing, legacy app backend)
05/2016 – 10/2016	**Mobile first Web application (React.js + node.js backend)**
www.mein-schulplan.de	Mobile and web applicatoin built on innovative stack using TFRP (react.js+eventemitter) in frontend and Node.js in backend.
Technologies	Node.js, React.js, CoffeeScript, PouchDB, CouchDB, koa
Tasks	➢ Design and prototyping of entire application stack ➢ Providing infrastructure for unit, integration and e2e tests
06/2015 – 12/2015	**Implementation of REST-API Stack**
Okotta GmbH *www.okotta.com*	Mobile App with REST data backend. Scalable data repositories secured with JSON Web Tokens.
Technologies	Scala, Scalatra, MongoDB, JWT, Swagger, WireMock, React.js, ES6

Tasks

- Requirements engineering and domain modeling
- Design and implementaion of REST-API stack
- Evaluation of persistence technologies
- Implementation of full stack integration testing
- Internal Data Tool based on ES6 and react.js

06/2014-12/2014 **REST API built on Google App Engine and Memcache**

Small Improvements
small-improvements.com

Web application running on GAE with BigTable and Memache. Angular frontend connects via Ajax to data backend.

Technologies Java 6, Groovy, GAE, Memcache, Google Guava, Jackson, Jmeter, Spock, Gradle

Tasks

- Performance optimization of object serialization
- (Re-)Implementation of REST-API with composable serializers
- Integration of caching layer using Guava Cache
- Refactoring to patterns

03/2013 - 09/2013 **Automated provisioning of Virtual Machines in Data Centers**

ProfitBricks GmbH
www.profitbricks.com

J2EE application handling customer requests for virtualized server infrastructure

Technologies Java 7, EJB 3, JPA 2 (EclipseLink), Postgres, Glassfish, jUnit, cglib, OSGI, H2, Jenkings, Maven, git

Tasks

- Rewrite test infrastructure with internal DSL and H2
- Lazy assertions with cglib and dynamic proxies
- Profiling and performance optimizations
- Code reviews and refactorings

02/2012 - 12/2012 **Web application with scalable relational social graph**

Alecto GmbH
www.friendsurance.de

Online platform offering comparison services around insurances and a social graph for customers to form trust networks

Technologies Java 7, EJB 3, JPA 2 (Hibernate), Spring 3, Postgres, Tomcat 7, Glassfish, Selenium, JSF, Maven, git

Tasks

- Full migration from EJB 3 (Glassfish) to Spring (Tomcat)
- Redesign of persistence abstraction layer
- Historized social graph based on immutable data (Postgres)
- Caching layer with synchronization across machines
- Selenium test infrastructure on grid server

Co-operative course of studies

15.10.2007 – 30.09.2008 **Softwaredevelopment ottogroup retail systems**

www.ottogroup.com

Support the migration of the ORM layer in a large J2EE landscape

Technologies Java 6, JDO 2.0, JPA 1.0, jUnit, Spring, Xtext, Xtend,Xpand,

Tasks

- Extension of existing persistence infrastructure for operating JDO and JPA in parallel (soft migration strategy)
- Automatic transformation of all OR-mapping definitions as part of my diploma thesis

Degree program

01.10.2004 –30.09.2008 **Co-operative degree program 'Economics and computer science (Dipl. FH)' in co-operation with ottogroup**

Focue areas Model driven software development ■ Modern software technologies ■ Datastream Managementsystems

Diploma thesis Implementation of a generic transformation component for XML based models and its application to a model migration problem

Final grades 1.0 (thesis), 1.5 (diploma)

IT-Qualifications

- 8+ years software development experience in J2EE business environment
- Strong experience running J2EE technologies in production: EJB, Spring, JTA, JPA, Servlet, JSP, application servers, servlet containers
- Object relationial persistence: Data and object modeling, query optimization, Postgres, MySQL
- Good understanding of architecture and design patterns: SOLID, Messaging, Pipes&Filters, REST, GoF Patterns
- Experience with model driven software development and modeling: Xtext, Xtend, Xpand, EMF, UML, AndroMDA
- Solid experience with XML technologies: XSD, XSLT, XPath
- Understanding of functional programming (Lisp/Scheme)
- Development Tools: Maven, Git, Ant, SVN,CVS,Jenkins
- Experienced Linux user

Open Source Projects

[github:mbassador] High-Performance Java Event Bus library: Implementation of publish-subscribe pattern. Low memory footprint, nonblocking iterators, high throughput

[github:mailody] Online trash mail service built with Node.js, MongoDB, restify, knockout.js, HTML 5, Java client for Selenium

[github:kermit] Kermit is an elegant, stream-based, extensible web scraper system. It is packed with pluggable features and offers a plugin mechanism for easy extension. Written in Node.js and coffeescript with NeDB as datastore.

Languages

*Foreign**

	oral	writing
English	C1	C2
Spanish	B2	C1
French	A1	A2

** Self evaluation according to refernce framework of European Language Portfolio (see appendix or www.coe.int/portfolio)*

Native Deutsch